

Re: drone reference design V2

From Loida Canada - A20687 <Loida.Canada@microchip.com>

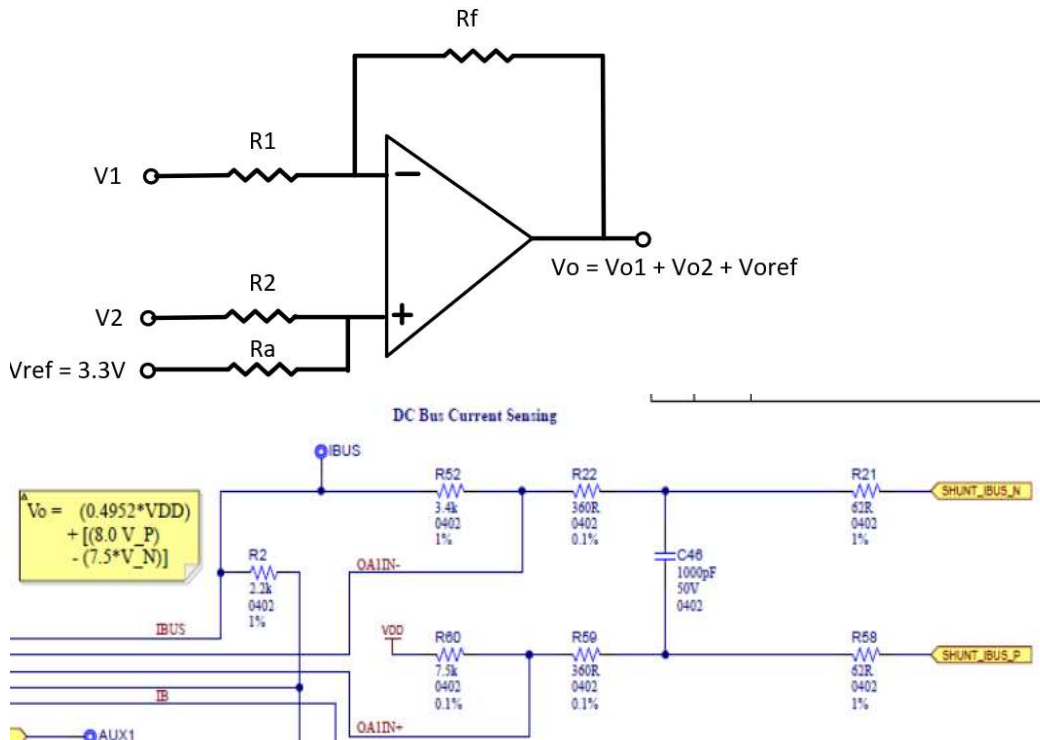
Date Thu 5/7/2026 8:12 AM

To Debraj Deb - I00235 <Debraj.Deb@microchip.com>; Severino Balong - A19614 <Severino.Balong@microchip.com>; Christoph Baumgartner - M91110 <Christoph.Baumgartner@microchip.com>; Srikar Deshmukh - I14317 <Srikar.Deshmukh@microchip.com>; Mike Gomez - A15693 <Mike.Gomez@microchip.com>; Jenny Puthusseri - I14883 <Jenny.Puthusseri@microchip.com>; Arjun TM - I66575 <ARJUN.TM@microchip.com>

Hi Debraj,

Good day!

I am following up on this topic to provide an updated calculation for the feedback resistor based on the series resistor value of 7.5 kΩ with VDD=VREF=3.3 V. Please refer to the calculation below:



$$R_f = \frac{1.65 (R_a + R_1)}{3.3} - R_1$$

$$R_f = \frac{1.65 (7500 + 422)}{3.3} - 422$$

$$R_f \approx 3539 \Omega$$

- The nearest available standard value in the market is:

- [ERA-2AEB3571X](#) = 3570 ohms
- Impact on System Parameters:

$$\text{Gain} = \frac{R_f}{R_{in}} = \frac{3570}{422} = 8.46$$

- $$I_{\max} = \frac{1.65}{R_{sh} \cdot \text{Gain}} = \frac{1.65}{0.01 \cdot 8.46} \approx 19.5 \text{ A}$$
- **Maximum Current:**
- **Output Equation:**
$$V_o = (0.5039 \cdot V_{DD}) + [(8.96 \cdot V_P) - (8.46 \cdot V_N)]$$

Given this modification, we plan to proceed with the procurement of the replacement resistor. We are sharing these updated calculations for your review, and we welcome any verification or feedback from your end.

Best regards,
Loida

From: Debraj Deb - I00235 <Debraj.Deb@microchip.com>

Sent: Thursday, April 30, 2026 12:00 AM

To: Severino Balong - A19614 <Severino.Balong@microchip.com>; Loida Canada - A20687

<Loida.Canada@microchip.com>; Christoph Baumgartner - M91110

<Christoph.Baumgartner@microchip.com>; Srikar Deshmukh - I14317

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<ARJUN.TM@microchip.com>

Subject: RE: drone reference design V2

Thank you Loida and Sev for the investigation and arriving at a probable root cause. Regarding the replacement of resistor that Sev is asking, I will have someone address it.

But once we correct the boards we must test them and only then ship any remaining boards. I am also in the opinion of sending the corrected boards to Christoph, but I am not sure if his customer will wait that long or if he can make those changes on his boards himself.

I will wait for Christoph to comment.

Regards,
Debraj

From: Severino Balong - A19614 <Severino.Balong@microchip.com>

Sent: Wednesday, April 29, 2026 6:27 PM

To: Loida Canada - A20687 <Loida.Canada@microchip.com>; Debraj Deb - I00235 <Debraj.Deb@microchip.com>;

Christoph Baumgartner - M91110 <Christoph.Baumgartner@microchip.com>; Srikar Deshmukh - I14317

<Srikar.Deshmukh@microchip.com>; Mike Gomez - A15693 <Mike.Gomez@microchip.com>; Jenny Puthusseri - I14883 <Jenny.Puthusseri@microchip.com>; Arjun TM - I66575 <ARJUN.TM@microchip.com>

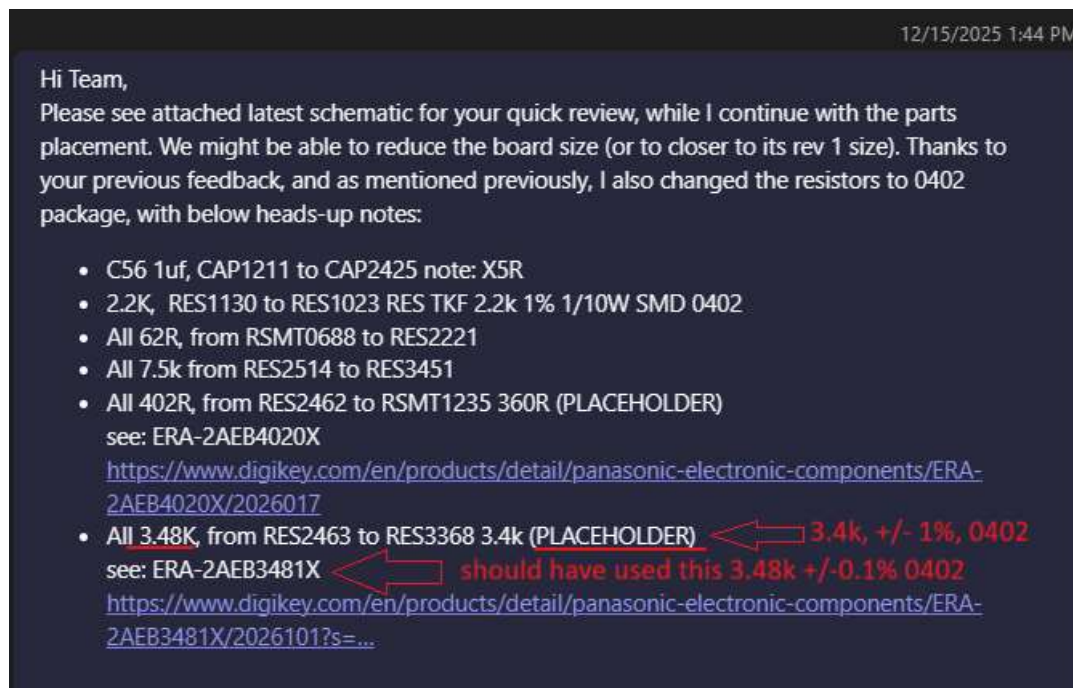
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Hi Loida,

Thank you for including me in the loop. I would like to provide additional context and further add details on the root cause.

From our separate discussions earlier today, and in addition to what we found out about the tolerance issue, we might also consider checking into the precise intended value which was 3.48 kΩ +/-0.1%, whereas the schematic currently reflects 3.4 kΩ +/-1%.

The 3.4 kΩ +/-1% value was initially used as a temporary placeholder to allow schematic reviews and PCB layout activities to proceed while the MCL Request was in progress to add the correct component to the Altium database.



Once the correct part was approved and added to the MCL database, the placeholder component (MCHPID: [RES3368](#), 3.4 kΩ +/-1%) should have been replaced with (MCHPID: [RES3551](#), 3.48 kΩ +/-0.1%).

Unfortunately, this replacement did not occur as intended.

Please confirm if replacing R52, R85 and R69, with MCHPID: RES3551, 3.48 kΩ +/-0.1% will be our next course of action and will update and push the Altium files to the repo, and initiate purchase.

I agree with your statement regarding taking full responsibility for this oversight, and I also apologize for any inconvenience caused.

Regards,
Sev

From: Loida Canada - A20687 <Loida.Canada@microchip.com>

Sent: Wednesday, April 29, 2026 3:11 PM

To: Debraj Deb - I00235 <Debraj.Deb@microchip.com>; Christoph Baumgartner - M91110 <Christoph.Baumgartner@microchip.com>; Srikar Deshmukh - I14317 <Srikar.Deshmukh@microchip.com>; Mike Gomez - A15693 <Mike.Gomez@microchip.com>; Jenny Puthusseri - I14883 <Jenny.Puthusseri@microchip.com>; Arjun TM - I66575 <ARJUN.TM@microchip.com>

Cc: Severino Balong - A19614 <Severino.Balong@microchip.com>

Subject: Re: drone reference design V2

Hi Debraj,

We have completed a review of the design history and schematics and have identified the root cause of the current sensing offset.

The issue stems from the feedback resistor tolerance being specified at 1% instead of 0.1%, which can introduce a larger offset in the current sensing stage. This was an oversight during the schematic update when transitioning from the 0603 to the 0402 package.

In terms of impact and mitigation:

- **User Guide:** The document is still under TPT review, so we can incorporate the correction directly without disruption to release timelines.
- **Hardware:** We will proceed with procuring and updating the design to use a 0.1% tolerance resistor to ensure improved accuracy in the current sensing stage.

We take full ownership of this oversight and are addressing it promptly to minimize downstream impact.

Please let us know if you would like us to include additional validation data or further analysis.

Best regards,
Loida

From: Debraj Deb - I00235 <Debraj.Deb@microchip.com>

Sent: Wednesday, April 29, 2026 10:37 AM

To: Christoph Baumgartner - M91110 <Christoph.Baumgartner@microchip.com>; Srikar Deshmukh - I14317 <Srikar.Deshmukh@microchip.com>; Loida Canada - A20687 <Loida.Canada@microchip.com>; Mike Gomez - A15693 <Mike.Gomez@microchip.com>; Jenny Puthusseri - I14883 <Jenny.Puthusseri@microchip.com>; Arjun TM - I66575 <ARJUN.TM@microchip.com>

Subject: RE: drone reference design V2

Hello Loida and Mike,

I am acting as devils advocate:

The boards which were sent to Christoph were the initial builds with manual soldering. Do you think that some resistors on the boards may have been wrongly soldered by the technician? Or the resistors used were of wider tolerances (1% instead of 0.1%?)

Regards,

Debraj

From: Christoph Baumgartner - M91110 <Christoph.Baumgartner@microchip.com>
Sent: Tuesday, April 28, 2026 8:58 PM
To: Srikar Deshmukh - I14317 <Srikar.Deshmukh@microchip.com>; Debraj Deb - I00235 <Debraj.Deb@microchip.com>; Loida Canada - A20687 <Loida.Canada@microchip.com>; Mike Gomez - A15693 <Mike.Gomez@microchip.com>
Subject: drone reference design V2

Hi Srikar, Debraj,

1000 thanks for Srikars outstanding support for this project.

I finally got all 6 pcbs I have received working.

Here are my findings.

The received board support file for motorBench required some updates.

If the resistors from the schematics for the voltage dividers 1.8k/18k (supply voltage, phase voltage) are correct the board support file requires some changes.

Due to the offset voltage generation with resistors I saw big and very different offsets for the 2 phase currents and on some boards the offset calibration wasn't working with the default setting of 3.125%.

Upgrading this value to 6.25% solved the issue on my six boards.

If the offset calibration fails the motor tries to start, but fails.

Because of my 24V requirement I also had to change the dcLink values in the board support file.

For my use case but also in general with the specs for the board the 25V rating of the cap is tight from my point of view.

On one board a UART pin on the MCU required soldering for a robust connection.

The DClink shunt resistor even at 1.2A reached more than 65°C.

The remaining issues are 2 possible fault conditions after a power cycle of the board

MCP802X_MOSFET_OVERCURRENT_FAULT = 16u,

or

MCP802X_MOSFET_UNDERVOLTAGE_LOCKOUT_FAULT = 32u,

What would be a good strategy if I have to clear this in the code?

I have created a custom GUI that clears the errors before starting, so I have a good workaround for the customer.

Still I want to find the root cause. Hope I can get one more bug fixing session with Srikar?.

Kind regards

Christoph